

Capacity of the Universe on the Basis of Observer-Function Analysis in Nonlocal Unification

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Abstract

We develop an information-theoretic extension of Nonlocal Unification (NU) in which the observer function is interpreted as a communication channel from fundamental informational change to perceived spacetime. The input to this channel is measured explicitly in bits, while the output is not identified with an assumed ontological spacetime but with an observer-dependent perceived spacetime configuration. Since every such configuration is a statistical object, its representational entropy can be measured in bits. We therefore define the capacity of an observer-function channel as the maximum mutual information between informational input and perceived spacetime output, and define a universal capacity by an appropriate supremum over observers.

We then introduce noisy observers by allowing the observer function itself to fluctuate. The resulting conditional entropy quantifies uncertainty in perceived spacetime induced specifically by perturbations of the representation function. The principal result of this paper is a supercapacity defect theorem. For a specified observer-function channel satisfying the strong converse, if the informational sampling rate exceeds that channel's capacity, then the probability of a fixed-tolerance perceived-spacetime reconstruction defect cannot be made arbitrarily small by increasing the number of informational channel uses. Under the strong converse, the corresponding block defect probability approaches unity. The theorem is formulated without a primitive transmission duration. Within NU this is significant because coordinate time is itself generated by observer functions from accumulated information. Consequently, duration cannot be invoked as an independent resource at the foundational level.

We discuss the quantum interpretation of this construction in light of the NU treatment of Hardy's informational reconstruction of quantum mechanics. NU retains Hardy's operational scaffolding while proposing information conservation as an additional principle. The present capacity construction is compatible with that informational perspective, but the universal capacity derived here should not be confused with a numerical bound on the entropy of the observable universe until the channel, its alphabet, and its noise model have been physically specified.

This work is dedicated to Albert Einstein.

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1 Introduction

Nonlocal Unification (NU) begins from the proposition that information is fundamental and conserved and seeks to use informational structure as a common language for quantum phenomena, classical measurement, consciousness, and spacetime. The published formulation explicitly places the theory in dialogue with informational reconstructions of quantum mechanics, including Hardy’s reconstruction, and proposes Nonlocal Consciousness (NC) as a global informational limiting structure toward which observer-dependent descriptions converge [1].

A central ingredient of NU is the observer function. In its basic phenomenological form,

$$\Delta t = f_H(\Delta I), \tag{1}$$

where ΔI is the information gained by an observer and f_H is the observer-specific mapping into temporal experience. The published theory subsequently generalizes this construction to observer-dependent spacetime mappings and explicitly treats spacetime intervals as emergent constructs of informational processing [1].

The purpose of the present work is to reinterpret this observer function as a communication channel. The input alphabet consists of changes in self-information, measured in bits. The output is not assumed to be “spacetime itself.” Rather, the output is the spacetime configuration available to an observer. This distinction is essential. An observer does not encounter an ontologically unmediated spacetime object; the observer constructs or experiences a representation of spacetime through a particular informational mapping.

We therefore write

$$I \longrightarrow f_H \longrightarrow X_H, \tag{2}$$

where I is informational input and X_H is a perceived spacetime configuration.

The central question is then:

How many informational bits can be represented reliably as perceived spacetime?

(3)

The answer is naturally expressed using Shannon mutual information and channel capacity.

The novel feature is that the channel itself is observer-dependent and can be noisy. We therefore consider

$$X_H = f_H(I, \Theta_H), \quad (4)$$

where Θ_H describes the state of the observer function and may fluctuate.

This leads to a natural separation between information present at the input and information successfully represented at the output.

2 NU as an Information-to-Perceived-Spacetime Channel

The published NU framework treats information as a fundamental conserved quantity and introduces Local Consciousness Instruments (LCIs) whose observer functions map informational gain into observer-relative temporal experience. Different LCIs may possess different functions, while sequences of observer functions are proposed to converge toward the NC limit [1].

The present construction makes one additional interpretive step. Instead of regarding f_H only as a map from information to subjective time, we regard it as the core of a more general map

$$f_H : \mathcal{I} \longrightarrow \mathcal{X}_H, \quad (5)$$

where $\mathcal{I}$ is an informational state space and $\mathcal{X}_H$ is the space of configurations that observer H can represent.

A configuration may contain perceived coordinates, intervals, metric components, fields, or other observer-accessible quantities:

$$X_H = \left(x_H^\mu, g_{\mu\nu}^{(H)}, \phi_H, \dots \right). \quad (6)$$

The exact content of X_H is deliberately left open at this stage. The essential requirement is that distinguishable configurations form a statistical ensemble.

For a discrete configuration space,

$$H(X_H) = - \sum_x p_H(x) \log_2 p_H(x) \quad (7)$$

is the entropy of perceived spacetime in bits.

This quantity should be distinguished from transmitted information. The entropy $H(X_H)$ measures the uncertainty or representational variety of the output, whereas

$$I(I; X_H) = H(X_H) - H(X_H|I) \quad (8)$$

measures the information about the input that survives in the perceived spacetime representation.

Thus the channel capacity of observer H is defined by

$$C_H = \sup_{p(I)} I(I; X_H). \quad (9)$$

This is a capacity in bits per fundamental channel use, not initially in bits per second.

3 Notation and Definitions

The notation used in the theorem and its proof is summarized in Table 1.

Table 1: Notation used in the observer-function capacity construction.

Symbol	Meaning
I	Fundamental informational input, measured in bits.
I^n	Block of n informational channel uses.
H	Shannon entropy in bits.
$I(I; X)$	Mutual information between input and perceived output.
f_H	Observer function associated with observer H .
Θ_H	Random state or parameters of the observer function.
X_H	Perceived spacetime configuration generated by observer H .
$\mathcal{X}_H$	Configuration space of perceived spacetime for H .
C_H	Capacity of the ideal observer-function channel.
C_H^N	Capacity of a noisy observer-function channel.
C_U	Universal communication capacity, defined over the observer family.
R	Informational sampling rate, in bits per channel use.
n	Number of fundamental informational channel uses.
$P_{\text{def}}^{(n)}$	Probability of a perceived-spacetime defect for a block of length n .
$\mathcal{E}_n$	Event that the reconstructed perceived configuration is defective.
δ	Tolerance defining acceptable configuration error.
Σ_Θ	Covariance of observer-function perturbations.
J_f	Jacobian of the observer function with respect to its parameters.
NC	Nonlocal Consciousness, the proposed global informational limit.

Definition 3.1 (Perceived-spacetime configuration). *A perceived-spacetime configuration is an observer-dependent statistical representation $X_H \in \mathcal{X}_H$ generated from informational input by the observer function f_H .*

Definition 3.2 (Perceived-spacetime defect). *Let $d_{\mathcal{X}}$ be a metric or distinguishability measure on $\mathcal{X}_H$. Given tolerance $\delta > 0$, define the defect event*

$$\mathcal{E}_n = \left\{ d_{\mathcal{X}}(X^n, \hat{X}^n) > \delta \right\}, \quad (10)$$

where $\hat{X}^n$ is the reconstructed perceived-spacetime configuration.

The perceived-spacetime defect probability is

$$P_{\text{def}}^{(n)} = \Pr(\mathcal{E}_n). \quad (11)$$

Definition 3.3 (Noisy observer). *A noisy observer is one for which the observer function is itself stochastic:*

$$X_H = f_H(I, \Theta_H), \quad (12)$$

where Θ_H is a random observer-function state.

4 Observer-Function Noise

The introduction of observer noise is important because ordinary communication theory commonly places noise in the physical transmission medium or at the output. Here the representation function itself can fluctuate.

Let

$$\Theta_H = \bar{\Theta}_H + \delta\Theta_H, \quad \mathbb{E}[\delta\Theta_H] = 0. \quad (13)$$

For small perturbations,

$$f_H(I, \Theta_H) \simeq f_H(I, \bar{\Theta}_H) + J_f(I)\delta\Theta_H, \quad (14)$$

where

$$J_f(I) = \left. \frac{\partial f_H(I, \Theta_H)}{\partial \Theta_H} \right|_{\bar{\Theta}_H}. \quad (15)$$

Consequently,

$$\Sigma_X(I) \simeq J_f(I)\Sigma_\Theta J_f(I)^T. \quad (16)$$

Thus observer-function noise induces uncertainty directly in perceived spacetime.

The corresponding conditional entropy

$$H(X_H|I) \quad (17)$$

has a direct interpretation: it is the uncertainty in perceived spacetime that remains even after the informational input is fixed.

Accordingly,

$$I(I; X_H) = H(X_H) - H(X_H|I) \quad (18)$$

quantifies the information about the fundamental informational state actually represented by the observer.

For the canonical NU candidate

$$f(\Delta I) = \sinh^{-1}(\alpha\Delta I), \quad (19)$$

a perturbation $\alpha \rightarrow \alpha + \delta\alpha$ gives

$$\frac{\partial f}{\partial \alpha} = \frac{\Delta I}{\sqrt{1 + \alpha^2 \Delta I^2}}, \quad (20)$$

and hence

$$\text{Var}(X|I) \simeq \frac{\Delta I^2}{1 + \alpha^2 \Delta I^2} \sigma_\alpha^2. \quad (21)$$

The published NU work selects the inverse hyperbolic sine as its preferred observer function because it is globally defined, monotonic, invertible, smooth, and exhibits saturation behavior; it also associates this choice with a square-root deformation of the canonical commutator [1].

5 Universal Capacity

Let $\mathcal{H}$ denote a family of physically admissible observers. For each observer,

$$C_H = \sup_{p(I)} I(I; X_H). \quad (22)$$

A natural observer-family capacity is

$$C_{\mathcal{H}} = \sup_{H \in \mathcal{H}} C_H. \quad (23)$$

We define the universal capacity by

$$C_U = \sup_{H \in \mathcal{H}} \sup_{p(I)} I(I; X_H). \quad (24)$$

An alternative joint-observer definition may be useful in future work:

$$C_U^{\text{joint}} = \sup_{p(I)} I(I; \{X_H\}_{H \in \mathcal{H}}). \quad (25)$$

The present theorem uses the single-channel quantity C_U .

The word “universal” here should therefore be interpreted carefully. Equation (24) does not yet claim a numerical capacity for the observable universe. It defines a capacity of the information-to-perceived-spacetime architecture once the physically admissible observer family and channel law have been specified.

6 The First Capacity Theorem

We now state the principal result.

Theorem 6.1 (Supercapacity perceived-spacetime defect theorem). *Consider the noisy observer-function channel*

$$I^n \longrightarrow X_H^n \quad (26)$$

with universal capacity

$$C_U = \sup_{H \in \mathcal{H}} \sup_{p(I)} I(I; X_H). \quad (27)$$

Assume that the selected channel is information-stable and satisfies the strong converse property. Let a sequence of codes sample the informational input at rate

$$R > C_U \quad (28)$$

bits per fundamental informational channel use. Then, for every $\epsilon < 1$, there exists n_0 such that for all $n > n_0$,

$$P_{\text{def}}^{(n)} > \epsilon. \quad (29)$$

Equivalently,

$$\boxed{\liminf_{n \rightarrow \infty} P_{\text{def}}^{(n)} \geq \epsilon_R > 0} \quad (30)$$

for some $\epsilon_R > 0$.

Under the strong converse in its usual form,

$$\boxed{\lim_{n \rightarrow \infty} P_{\text{def}}^{(n)} = 1.} \quad (31)$$

Therefore increasing the number of informational channel uses cannot restore arbitrarily reliable perceived-spacetime representation when the input sampling rate exceeds C_U .

6.1 Proof

For a fixed observer channel, let C_H denote its capacity. By definition of C_U ,

$$C_H \leq C_U \quad (32)$$

for every admissible observer H .

Suppose a code operates at rate

$$R > C_U. \quad (33)$$

Then necessarily

$$R > C_H \quad (34)$$

for every admissible H .

The standard coding theorem states that rates below capacity can admit sequences of codes whose error probability tends to zero. The converse states that reliable communication is impossible above capacity. The strong converse strengthens this statement: for a channel satisfying the strong converse, any sequence of codes whose rate exceeds capacity has error probability tending to one.

Let $\hat{X}^n$ denote the reconstructed perceived-spacetime configuration. By definition, the perceived-spacetime defect event is

$$\mathcal{E}_n = \left\{ d_{\mathcal{X}}(X^n, \hat{X}^n) > \delta \right\}. \quad (35)$$

The decoder therefore constitutes a communication code for the perceived-spacetime representation. Its block defect probability is

$$P_{\text{def}}^{(n)} = \Pr(\mathcal{E}_n). \quad (36)$$

Since $R > C_H$, the strong converse gives

$$\lim_{n \rightarrow \infty} P_{\text{def}}^{(n)} = 1. \quad (37)$$

Consequently, for every $\epsilon < 1$, there exists n_0 such that

$$P_{\text{def}}^{(n)} > \epsilon \quad (38)$$

for every $n > n_0$.

This establishes Eq. (29) and the stronger limit Eq. (31). $\square$

7 Why Transmission Duration Is Not a Primitive Variable

The theorem has been deliberately stated in terms of the number n of fundamental informational channel uses rather than a transmission duration T .

This distinction is central to the NU interpretation.

In conventional communication theory one frequently writes a capacity in units of bits per second and obtains a total transmittable information of the form

$$N_{\text{max}} = CT. \quad (39)$$

Such a statement assumes that temporal duration is already available as an independent parameter of the communication process.

The NU construction instead starts from informational accumulation. The published formulation writes

$$t = f(I), \quad (40)$$

and uses accumulated information as the dynamical parameter replacing coordinate time in the informational formulation [1].

The foundational ordering is consequently

$$\boxed{I \longrightarrow f_H \longrightarrow X_H \longrightarrow t_H.} \quad (41)$$

Time is therefore an output or derived coordinate of the observer-function construction rather than an independent parameter assumed at the input.

At the present stage, the meaningful sampling rate is consequently

$$R = \frac{\text{input information}}{\text{fundamental informational channel uses}}, \quad (42)$$

not

$$\frac{\text{bits}}{\text{seconds}}. \quad (43)$$

The theorem therefore does not say that transmission over an arbitrarily long physical time fails. Rather, it says something logically prior: before an observer-dependent temporal coordinate

has been generated, there is no primitive temporal duration that can be invoked to cure a supercapacity representation.

If

$$R > C_U, \quad (44)$$

then after n informational channel uses the attempted information is

$$N_{\text{in}} = nR, \quad (45)$$

whereas the reliable representational capacity scales as

$$N_{\text{reliable}} \lesssim nC_U. \quad (46)$$

The excess informational demand is therefore

$$N_{\text{excess}} = n(R - C_U). \quad (47)$$

Increasing n increases both quantities while preserving their rate difference. It does not convert a supercapacity channel into a subcapacity channel.

Thus the relevant question is not “how long was the information transmitted?” but rather “was the informational sampling density within the representational capacity of the observer-function channel?”

8 Quantum Aspects

The proposed capacity construction is intended to remain within the informational orientation of NU rather than replace its quantum foundations.

The NU paper explicitly discusses Hardy’s reconstruction of quantum mechanics and lists five structural axioms concerning convexity, continuity, composition, reversibility, and dimensional consistency. It then proposes information conservation as an additional principle rather than as a replacement for those axioms [1].

The present construction is compatible with that strategy in the following sense. The input I need not be restricted to classical information. A quantum system may generate operationally accessible information through measurement, and the observer-function channel can then map that information into a classical perceived configuration.

Schematically,

$$\rho \longrightarrow \text{quantum measurement} \longrightarrow I \longrightarrow f_H \longrightarrow X_H. \quad (48)$$

The quantum state ρ therefore need not itself be identified with perceived spacetime. Rather, the observer-function channel describes the informational registration by which a quantum process becomes part of an observer-defined classical configuration.

For a quantum input, the classical mutual-information capacity can be replaced by an appropriate quantum-information quantity. For a quantum channel $\mathcal{N}$,

$$Q(\mathcal{N}) \quad (49)$$

may denote a quantum capacity, while

$$C_{\text{cq}} = \sup_{\{p_i, \rho_i\}} I(X; Y) \quad (50)$$

describes a classical communication capacity of a quantum-to-classical measurement channel.

The present paper does not claim that C_U is automatically equal to either a quantum capacity or a Holevo quantity. Such an identification requires an explicit quantum channel model.

What NU contributes is the possibility that the final classical output space is itself an emergent perceived geometry:

$$\rho \rightarrow I \rightarrow f_H \rightarrow X_H \rightarrow g_{\mu\nu}^{(H)}. \quad (51)$$

This is particularly relevant because the NU formulation associates the observer function with deformations of canonical quantum commutation relations. Its preferred function

$$f(\Delta I) = \sinh^{-1}(\alpha \Delta I) \quad (52)$$

is reported to produce a square-root commutator deformation [1].

Thus observer-function noise could, in principle, propagate through both the perceived geometry and the quantum sector. If

$$\Theta_H = \bar{\Theta}_H + \delta\Theta_H, \quad (53)$$

then

$$\delta X_H \simeq J_f \delta\Theta_H \quad (54)$$

and, through the emergent metric,

$$\delta g_{\mu\nu}^{(H)} \simeq \frac{\partial g_{\mu\nu}^{(H)}}{\partial f_H} \delta f_H. \quad (55)$$

This provides a possible bridge between observer-function noise, perceived-spacetime uncertainty, and quantum-geometric fluctuations.

The quantum interpretation should nevertheless remain appropriately qualified. The fact that NU is formulated within Hardy's informational framework means that the present construction is intended to be compatible with that operational architecture; it does not by itself prove a new quantum capacity bound.

9 Summary of the Proposed Framework

The discussion developed above can be summarized by a sequence of conceptual transformations. First, information is treated as the primitive input:

$$I \in \mathcal{I}. \quad (56)$$

Second, the observer function is promoted from a phenomenological mapping of information gain into subjective time to a communication channel:

$$f_H : \mathcal{I} \rightarrow \mathcal{X}_H. \quad (57)$$

Third, the output is explicitly interpreted as perceived spacetime rather than presumed ontological spacetime:

$$X_H \in \mathcal{X}_H. \quad (58)$$

Fourth, every representational configuration has an entropy,

$$H(X_H), \quad (59)$$

which provides an information-theoretic measure, in bits, of the configuration space available to the observer.

Fifth, the amount of input information represented in perceived spacetime is the mutual information

$$I(I; X_H) = H(X_H) - H(X_H|I). \quad (60)$$

Sixth, the capacity of the observer-function channel is

$$C_H = \sup_{p(I)} I(I; X_H). \quad (61)$$

Seventh, the universal capacity is defined by an appropriate supremum over admissible observer functions:

$$C_U = \sup_{H \in \mathcal{H}} \sup_{p(I)} I(I; X_H). \quad (62)$$

Eighth, observer noise is introduced at the most fundamental point by perturbing f itself:

$$f_H \rightarrow f_H(I, \Theta_H). \quad (63)$$

The resulting conditional entropy

$$H(X_H|I) \quad (64)$$

is interpreted as representation uncertainty induced by the noisy observer-function channel.

Finally, a perceived-spacetime defect is defined through a configuration distance,

$$d_{\mathcal{X}}(X, \hat{X}) > \delta. \quad (65)$$

The first theorem then states that an informational sampling rate above universal capacity cannot produce arbitrarily reliable perceived-spacetime representation. Under the strong converse,

$$R > C_U \implies P_{\text{def}}^{(n)} \rightarrow 1. \quad (66)$$

This result is the first bridge between NU observer-function analysis and a universal communication-capacity concept.

10 Discussion and Limitations

The result should not presently be interpreted as a numerical calculation of the number of bits contained in the universe. Three mathematical objects remain to be specified before such a calculation can be attempted.

First, the physically admissible informational alphabet or quantum informational state space must be defined. Second, the output configuration space $\mathcal{X}_H$ must be operationally specified. Third, the probability law describing perturbations of f_H must be determined.

There is also an important distinction between three quantities:

$$H(X_H), \tag{67}$$

which measures output representational entropy,

$$I(I; X_H), \tag{68}$$

which measures information transferred from input to output, and

$$C_H, \tag{69}$$

which is the maximum such transfer over admissible input distributions.

Conflating these quantities would lead to an incorrect estimate of universal capacity.

A second limitation concerns the strong converse. The theorem as stated requires an information-stable channel satisfying a strong converse. Capacity alone does not guarantee that the error probability converges to one above capacity for every mathematically conceivable channel. Therefore the theorem is intentionally conditional on standard strong-converse hypotheses.

A third limitation is conceptual. The statement that time is not fundamental is a claim of the NU framework, not a consequence of Shannon theory. Information theory supplies the supercapacity theorem; NU supplies the interpretation that temporal duration is generated by the observer function. The two claims should remain logically distinct.

The published NU paper itself describes the framework as interpretative and identifies empirical validation, specification of the observer function, and the operationalization of information across physical domains as open issues [1]. These limitations apply directly to the present capacity construction.

11 Conclusion

We have formulated an observer-function version of a communication channel in which fundamental informational change is the input and perceived spacetime is the output. This construction changes the question of universal capacity from a vague question about how many bits “exist” in the universe into a precise question about how many informational distinctions can be reliably represented through the observer-dependent mapping from information to perceived spacetime.

The central quantity is

$$\boxed{C_U = \sup_{H \in \mathcal{H}} \sup_{p(I)} I(I; X_H).} \quad (70)$$

For noisy observers,

$$X_H = f_H(I, \Theta_H), \quad (71)$$

and observer-function perturbations generate perceived-spacetime noise

$$\Sigma_X = J_f \Sigma_\Theta J_f^T. \quad (72)$$

The first theorem establishes the supercapacity consequence:

$$\boxed{R > C_U \implies P_{\text{def}}^{(n)} \not\rightarrow 0,} \quad (73)$$

and, under the strong converse,

$$\boxed{R > C_U \implies P_{\text{def}}^{(n)} \rightarrow 1.} \quad (74)$$

The significance for NU is that this result requires no primitive transmission duration. The fundamental variable is the number of informational channel uses. Temporal duration becomes meaningful only after the observer function generates a perceived temporal coordinate,

$$I \rightarrow f_H \rightarrow X_H \rightarrow t_H. \quad (75)$$

Consequently, increasing a putative transmission duration cannot rescue a supercapacity informational representation at the foundational level: duration is itself part of the output geometry whose representational capacity is being studied.

The proposed programme therefore replaces the conventional question

$$\text{“How many bits can be transmitted per second?”} \quad (76)$$

with the more primitive question

$$\boxed{\text{“How many informational distinctions can be reliably represented as perceived spacetime?”}} \quad (77)$$

If this construction can subsequently be grounded in the quantum informational structure of NU and its NC limit, it may provide a route toward a genuinely universal communication capacity, while retaining a clear distinction between established information-theoretic results and the new physical hypotheses of Nonlocal Unification.

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